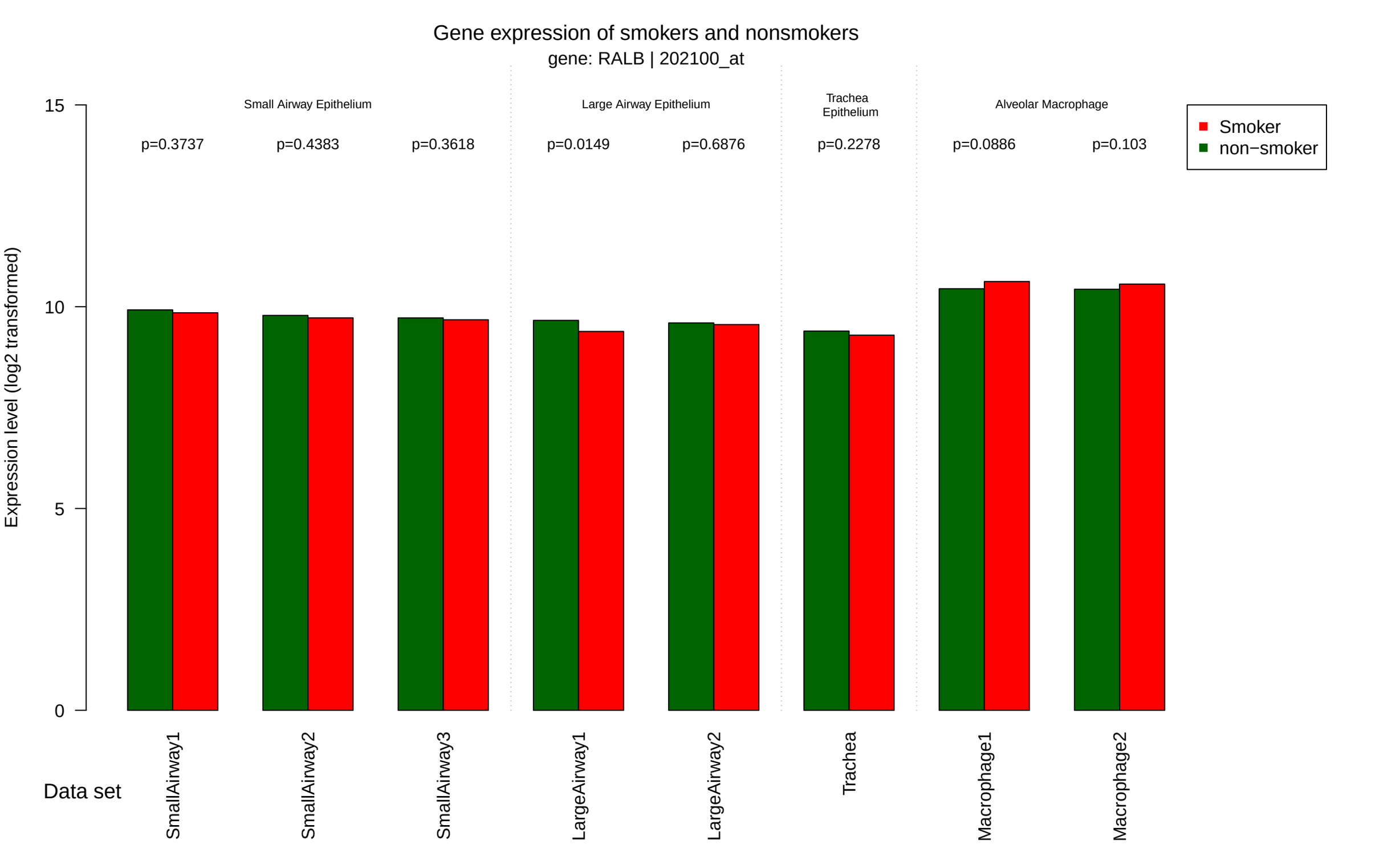
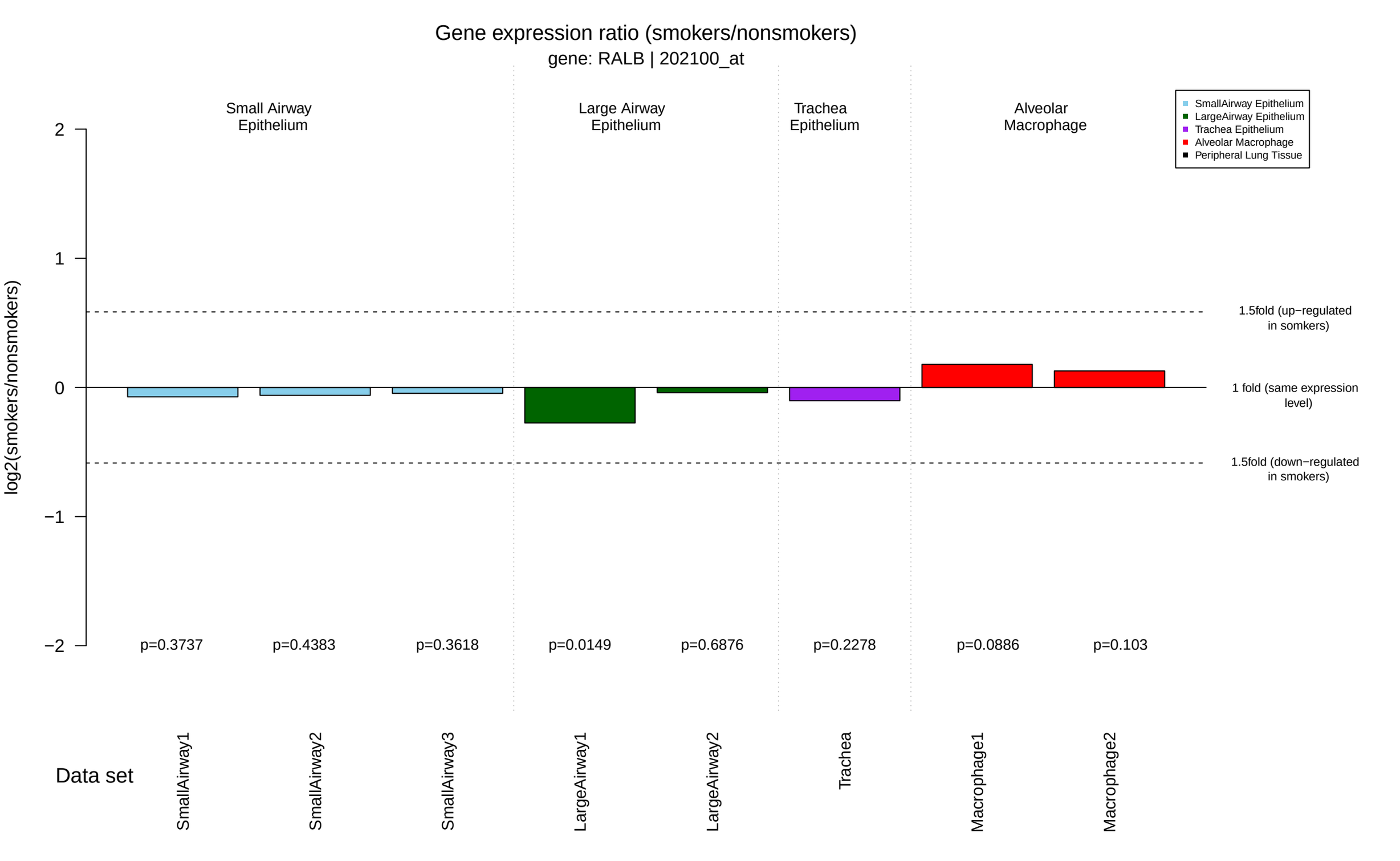
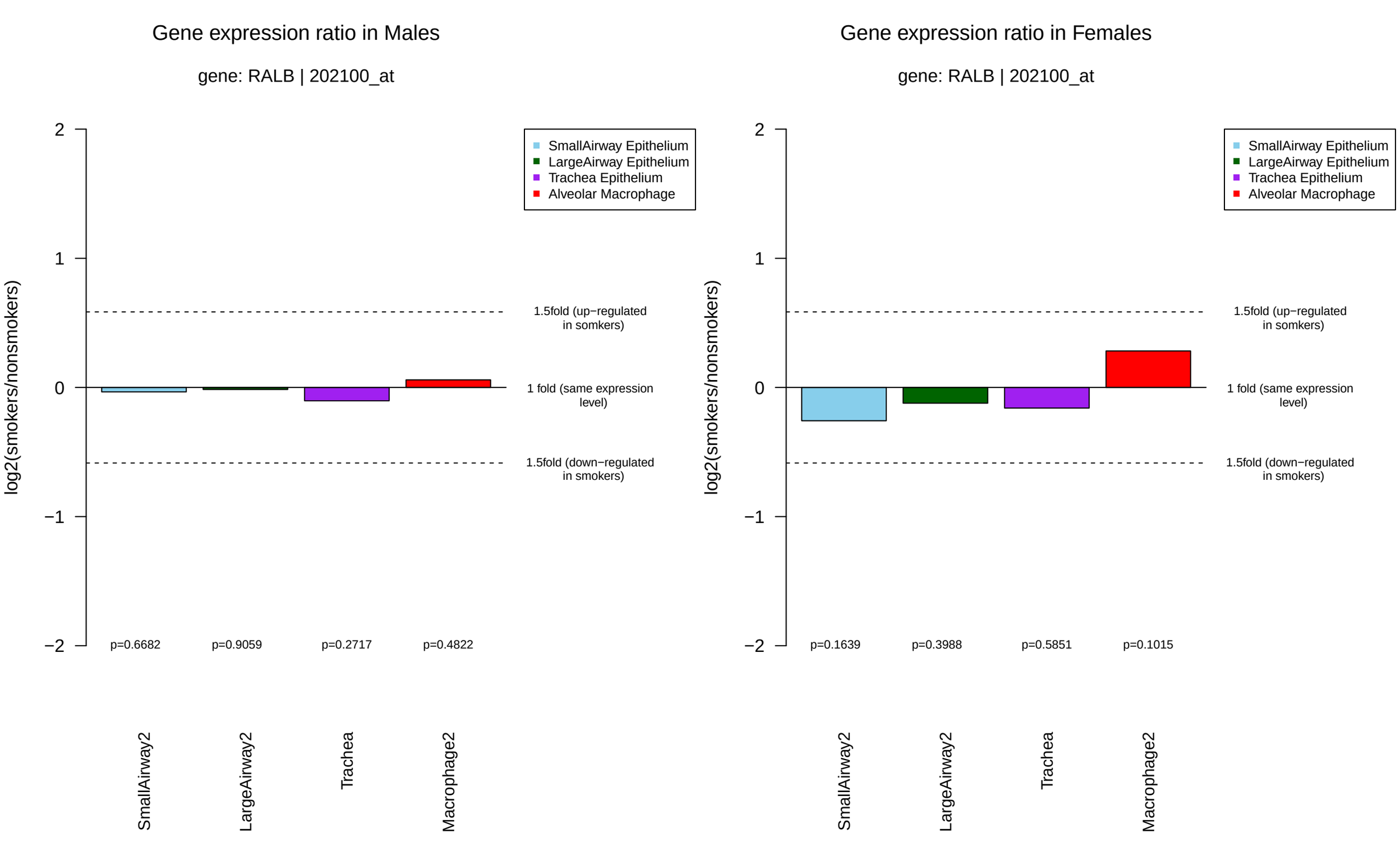
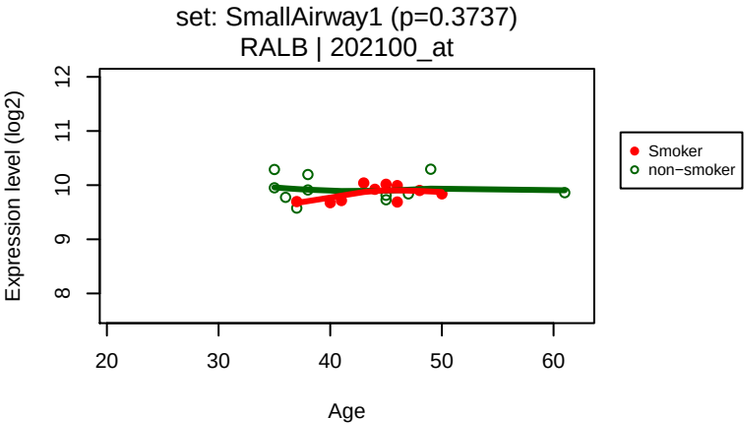
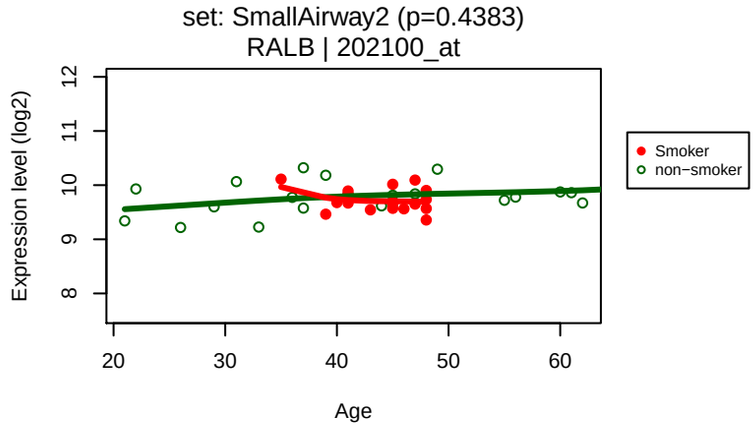
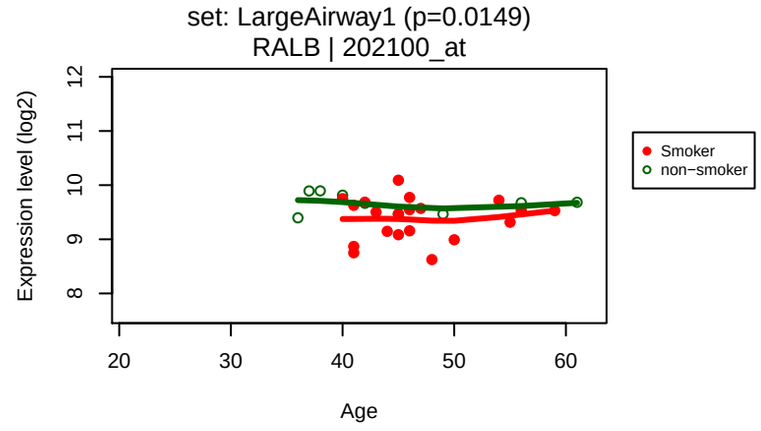

Gene expression of smokers and nonsmokers
gene: RALB | 202100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3737
p=0.4383
p=0.3618
p=0.0149
p=0.6876
p=0.2278
p=0.0886
p=0.103
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

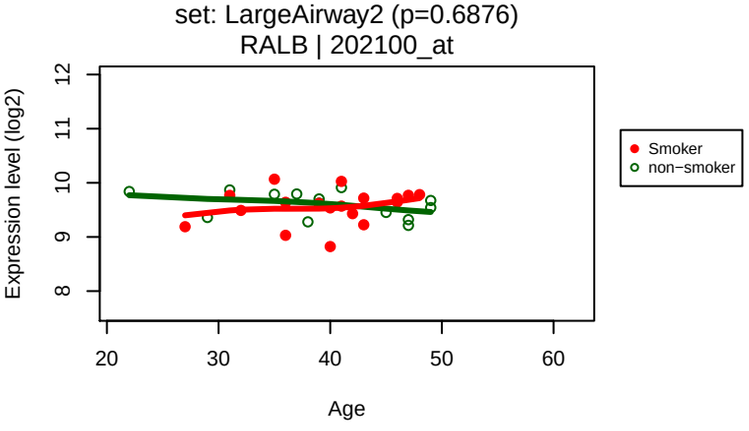

Gene expression ratio (smokers/nonsmokers)
gene: RALB | 202100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3737
p=0.4383
p=0.3618
p=0.0149
p=0.6876
p=0.2278
p=0.0886
p=0.103
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

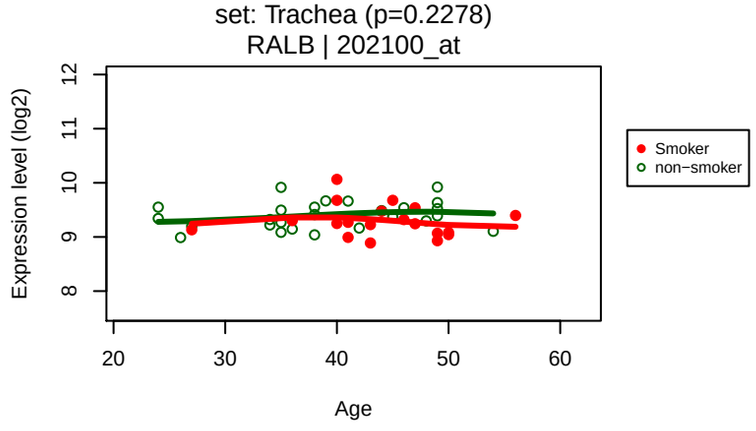

Gene expression ratio in Males
gene: RALB | 202100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6682
p=0.9059
p=0.2717
p=0.4822
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALB | 202100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1639
p=0.3988
p=0.5851
p=0.1015
SmallAirway2
LargeAirway2
Trachea
Macrophage2

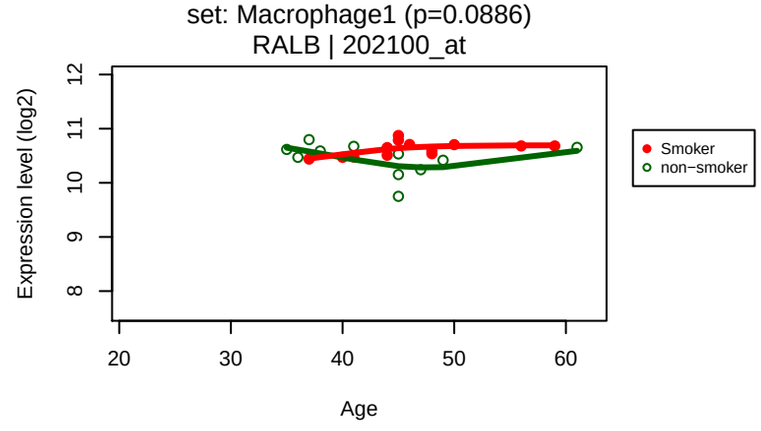

set: SmallAirway1 (p=0.3737)
RALB | 202100_at
Expression level (log2)
Age
Smoker
non−smoker

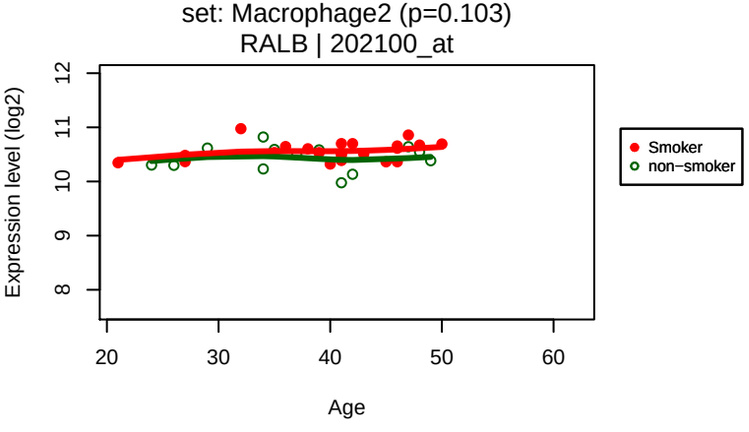

set: SmallAirway2 (p=0.4383)
RALB | 202100_at
Expression level (log2)
Age
Smoker
non−smoker

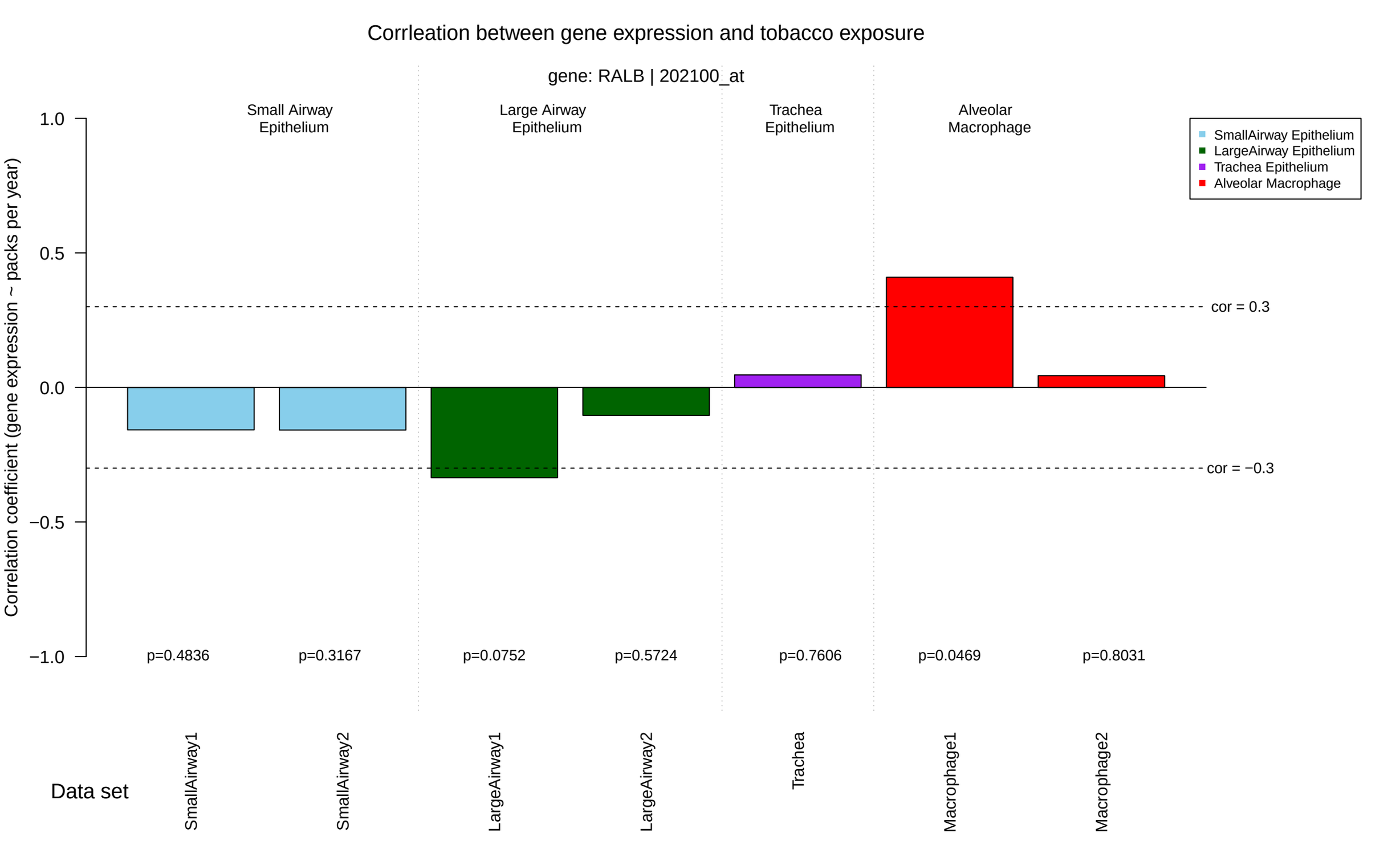

set: LargeAirway1 (p=0.0149)
RALB | 202100_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6876)
RALB | 202100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2278)
RALB | 202100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0886)
RALB | 202100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.103)
RALB | 202100_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALB | 202100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4836
p=0.3167
p=0.0752
p=0.5724
p=0.7606
p=0.0469
p=0.8031
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2